

An analysis of the four papers reveals how their methodologies align with **Problem Statement ID 26141 (Quantum-Inspired Cyber Threat Detection for Digital Signature Security)**, their mathematical parameters, their research gaps, and the techniques that can be adapted for your project.

1. Do They Match Our Project Core Idea?

Thematic & Terminological Match: Strong

- **Paper 4 (Abdullayev et al., 2026)** shares nearly the exact title prefix ("*Quantum-Inspired Cyber Threat Detection...*"). It formulates threat detection using quantum mathematical models (superposition, amplitude inner products, Grover amplification, and entanglement correlation) executed purely on classical hardware without quantum processors.
- **Paper 1 (Hussain et al., 2026)** integrates Quantum Key Distribution (QKD), QBER thresholds, and post-quantum digital signatures with blockchain.
- **Paper 2 (Somasundaram et al., 2026)** and **Paper 3 (Abdel-Aty et al., 2026)** focus on quantum-inspired threat detection and predictive defense.

Core Scope & Constraint Match: Significant Divergence

- **Target Domain:** Your problem statement focuses specifically on **Digital Signature Security in Teleportation-Based Quantum Digital Signature (QDS)** systems (detecting signature forgery, signer impersonation, replay attacks, and quantum channel tampering on Pauli-encoded states). In contrast, all four papers address **Network Intrusion Detection Systems (IDS) for IoT/IIoT network traffic flows** (detecting DoS, botnets, DDoS, port scans, and false data injection).
- **The "Zero-AI/ML" Constraint:** Problem Statement 26141 states that detection must function "**without relying on artificial intelligence or machine learning techniques**" to preserve **Information-Theoretic Security (ITS)**.
- Papers 1, 2, and 3 rely heavily on deep learning (CNNs, LSTMs, and Variational Quantum Classifiers).
- Paper 4 avoids neural networks and physical QPUs, using quantum linear algebra, but still relies on offline training against reference attack signatures.

2. Approaches Used in the Papers

Paper 4: QI-CTAS (Abdullayev et al., 2026)

- **Mathematical Vectorization:** Normalizes classical telemetry vectors into unit-norm quantum-inspired state vectors $|\psi\rangle$ satisfying the Born rule ($\sum |a_k|^2 = 1$).
- **State Overlap Scoring:** Evaluates threat levels via Hilbert-space inner products $|\langle \varphi_j | \psi_i \rangle|^2$ against signature states.
- **Grover-Inspired Amplitude Amplification:** Applies an iterative rotation operator to amplify anomalous components in heavily imbalanced data without neural networks.
- **Entanglement Correlation:** Uses a tensor-product-style correlation metric across nodes to detect coordinated lateral movement.

Paper 1: QASB-IDS (Hussain et al., 2026)

- **Hybrid Quantum-Classical Infrastructure:** Combines Decoy-State BB84 QKD for gateway session key exchange with NIST PQC (CRYSTALS-Dilithium) for signing blockchain transactions.
- **Consortium Blockchain Anchoring:** Employs a 3-tier Hierarchical PBFT blockchain to log intrusion alerts and model updates asynchronously.
- **Traffic Classification:** Uses an edge-quantized CNN-LSTM model with Differential Privacy (Gaussian mechanism) for collaborative federated learning.

Paper 2: QS-CDT (Somasundaram et al., 2026)

- **Cyber Digital Twin (CDT):** Replicates live network states (S_t) in a virtual sandbox, computing continuous deviation metrics $D(S_t, DT_t)$.
- **Quantum-Inspired Machine Learning (QML):** Uses Tensor Networks (Matrix Product States) and Variational Quantum Circuits (VQCs) simulated on classical hardware to classify encrypted metadata.
- **Automated SDN Orchestration:** Triggers dynamic OpenFlow table rules to isolate nodes and revoke PQC keys (CRYSTALS-Kyber/Dilithium).

Paper 3: Hybrid CNN-RNN-VQC (Abdel-Aty et al., 2026)

- **Spatiotemporal Feature Extraction:** CNN extracts spatial packet features; RNN/LSTM extracts temporal sequence dependencies.
- **VQC Classification:** Merged features are amplitude-encoded into a 4-to-6 qubit parameterized quantum circuit with R_x, R_y, R_z , and CNOT gates, optimized via the classical Adam optimizer.

3. Parameters Passed Across the Papers

Parameter Category	Specific Parameters & Values Reported	Source
Quantum & Cryptographic	$\text{QBER} \leq 5\%$ error tolerance; QKD key rate $R_s = 10 \text{ kbps} - 1 \text{ Mbps}$; NIST PQC Category 1 (Dilithium-II, Kyber-512); QKD Outage $P_{\text{out}} = \Pr(R_s < R_{\min})$; Session refresh $T_{\text{refresh}} = \frac{K_{\text{session}}}{R_s}$.	

| Hussain et al.

| | **Quantum-Inspired Scoring** | State normalization $\sum_{k=1}^n |a_{i,k}|^2 = 1$; Severity weights w_j ; Grover iterations $G = 3$; Amplification factor η ; Composite risk weights: $\delta_1 = 0.45$ (Threat Amplitude), $\delta_2 = 0.30$ (Correlation), $\delta_3 = 0.25$ (Trust); Trust decay floor $\text{Trust}_{\min}$.

| Abdullayev et al.

| | **Circuit & Deep Learning** | Qubits $N_q \in \{4, 6\}$; Learning rate $\alpha = 0.001$; Batch size $= 32 / 64$; Epochs $= 50 - 80$; Pruned parameter count $= 1.47 \text{ M}$ (\$1.82 MB memory footprint); DP noise scale $\sigma = 0.3$, privacy budget $\epsilon \approx 1.76$, $\delta = 10^{-5}$.

| Abdel-Aty et al., Hussain et al.

| | **Policy Enforcement** | Risk thresholds: Monitor ($\text{ARI} \leq 0.25$), Restrict ($0.25 < \text{ARI} \leq 0.45$), Quarantine ($0.45 < \text{ARI} \leq 0.70$), Isolate ($\text{ARI} > 0.70$).

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		IMMUTABLE AUDIT / KEY
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| | * Decoy-State QKD Key

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| | * NIST PQC Dilithium
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5. Research Gaps: What They Lack (Which Our Project Must Do)

1. Failure to Address Digital Signatures (QDS):

- None of the papers detect attacks on **teleportation-based digital signatures**. They treat security as general network intrusion detection.
- *What We Do*: Design the detection engine specifically around **teleportation-based QDS protocols**, verifying signature authenticity through Bell-state teleportation ($|\Phi^+\rangle_{AB}$) and Pauli unitary corrections ($I, \sigma_x, \sigma_y, \sigma_z$).

2. Violation of the "Zero-AI/ML" Constraint:

- Papers 1, 2, and 3 depend on deep neural networks (CNNs, LSTMs, VQCs). Neural networks are black boxes that make probabilistic mistakes (false alarms, evasion via adversarial perturbations), violating **Information-Theoretic Security (ITS)**.
- *What We Do*: Replace all neural networks with **rank-1 projective measurements** ($\Pi_{\text{pass}} = |\psi\rangle\langle\psi|$) and **Chernoff–Hoeffding statistical threshold bounds**, yielding exact mathematical proofs of security without training data.

3. No Representation of Pauli Eigenstates & Mutually Unbiased Bases:

- The papers encode generic classical numbers into state amplitudes. They do not model physical conjugate bases.
- *What We Do*: Encode message bits into the **6 Pauli eigenstates** across three mutually unbiased bases ($\sigma_x, \sigma_y, \sigma_z$). This guarantees that an eavesdropper or forger without prior basis knowledge introduces a detectable **$\geq 25\%$ to 33.3% error rate**.

4. Missing Quantum Channel Integrity Validation (Bell Test):

- None of the papers verify whether the quantum entanglement link itself is compromised before trusting the data.
- *What We Do*: Incorporate the **CHSH Bell inequality test** ($S \leq 2\sqrt{2}$) on decoy Bell pairs to detect intercept-resend attacks and channel decoherence before signature verification begins.

5. No Enforcement of Quantum State Destruction (Anti-Replay):

- None of the papers utilize quantum measurement collapse to prevent replay attacks.
- *What We Do*: Leverage the **No-Cloning Theorem** and projective measurement collapse—measuring the quantum public key physically destroys it, ensuring replay attacks fail deterministically ($P_{\text{detect}} = 1.0$).

6. Novel Techniques from These Papers We Can Apply to Our Project

While the papers address network flows and use AI, several mathematical and architectural techniques can be directly integrated into our QDS project:

1. Grover-Inspired Amplitude Amplification for Weak Channel Disturbance (from Paper 4):

- *Concept:* Paper 4 uses Grover rotations to amplify weak anomalies in imbalanced datasets.
- *Application to QDS:* In realistic fiber channels, an eavesdropper might only intercept a tiny fraction of transmitted qubits (a weak intercept attack). We can adapt the Grover operator:

$$a_j(t+1) = a_j(t) \cdot \left[1 + \eta (E_{\{\text{obs}, j\}} - \mu_E) \right]$$

to amplify trace measurement disturbances caused by subtle quantum channel manipulation before calculating the final QBER.

2. Entanglement-Inspired Correlation for Verifier Consensus (from Paper 4):

- *Concept:* Paper 4 models multi-node compromise via an overlap operator: $ECS_{\{i,i'\}} = \left| \langle \psi_i | \psi_{i'} \rangle \right|^2 \cdot C_{\{i,i'\}}$.
- *Application to QDS:* In multi-receiver QDS schemes (Alice signing a message verified by both Bob and Charlie), the protocol must prevent repudiation (Alice making Bob accept while Charlie rejects). We can adapt this overlap operator to compare the cross-verifier state fidelities between Bob and Charlie to verify multi-party signature consistency deterministically.

3. Asynchronous Consortium Blockchain Ledger for Nonce/Key Invalidation (from Paper 1):

- *Concept:* Paper 1 logs cryptographic updates to an asynchronous hierarchical PBFT blockchain to avoid network bottlenecks.
- *Application to QDS:* When Bob completes projective measurements on Alice's teleported state, the one-time Session ID (SID) and key index must be instantly burned. Writing consumed session IDs to a lightweight, permissioned ledger provides an immutable audit trail that prevents replay attacks across distributed nodes.

4. Adaptive Graduated Enforcement Ladder (from Paper 4 & Paper 2):

- *Concept:* Rather than a binary accept/reject flag, the system uses a graduated state machine: Monitor $\rightarrow$ Restrict $\rightarrow$ Quarantine $\rightarrow$ Isolate.

- *Application to QDS*: We can map our statistical QBER output into an automated response pipeline:
- $\text{QBER} \leq T_v$: Signature Accepted (Legitimate).
- $T_v < \text{QBER} < T_f$: Flag Channel Noise / Request Partial Retransmission on decoy states.
- $\text{QBER} \geq T_f$: Signature Rejected / Immediate Key Revocation / Alert Security Orchestrator (Forgery or Impersonation).

5. QKD Session Refresh & Outage Modeling (from Paper 1):

- *Concept:* Paper 1 calculates session key refresh time as $T_{\text{refresh}} = \frac{K_{\text{session}}}{R_s}$ and models link outage probability $P_{\text{out}} = \Pr(R_s < R_{\min})$.
- *Application to QDS:* We can adopt this formula in our simulator to dynamically calculate how many Bell pairs must be generated and distributed per second to sustain continuous digital signature operations under varying channel noise conditions.

Comparison Matrix: Literature vs. Our Solution

| Dimension | QASB-IDS (Paper 1) |

| QS-CDT (Paper 2)

| CNN-RNN-VQC (Paper 3)

| QI-CTAS (Paper 4)

 Our Proposed Framework (PS 26141) --- --- --- --- --- --- Primary Domain IIoT Intrusion Detection

| Network Threat Forecasting

| Network Anomaly Classification

| IoT Device Threat Detection

Teleportation-Based Quantum Digital Signatures (QDS)		Detection Method		CNN-LSTM Deep Learning	
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| Tensor Networks + QML

| CNN + RNN + VQC Classifier

| Quantum-Inspired Vector Overlap

| **Projective Measurements & Statistical Bounds (Zero-AI) | | Quantum Mechanism |**

Decoy BB84 QKD + Dilithium

| Kyber/Dilithium PQC

| VQC (Rotation & CNOT gates)

| Superposition math, Grover reweighting

| **6-State Pauli Eigenstates, Bell Pairs, Teleportation & CHSH | | Threats Targeted |** DoS,

FDI, MitM

| DDoS, Ransomware, Phishing

| Botnet, SQLi, PortScan

| Botnet recruitment, Replay

| **Forgery, Impersonation, Replay, Channel Tampering | | Security Proof |** Computational
/ Heuristic

| Computational / Heuristic

| Heuristic (Empirical accuracy)

| Empirical / Heuristic

| **Information-Theoretic Security ($P_{\text{forge}} \leq 2^{-cL}$) |**